

Ian McCormack

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EDUCATION

Ph.D. in Software Engineering, Carnegie Mellon University

September 2021 - Present

- Coadvised by [Jonathan Aldrich](#) and [Joshua Sunshine](#)
- [2024 CyLab Presidential Fellowship](#) 2024-2025
- National Science Foundation Graduate Research Fellow (NSF GRFP) 2021-2026

University of Wisconsin-Eau Claire

summa cum laude, May 2021

- **B.S.**, Computer Science
- **B.A.**, English, Rhetorics of Science, Technology, and Culture
- **Minor**, Mathematics

PUBLICATIONS

[A Mixed-Methods Study on the Implications of Unsafe Rust for Interoperation, Encapsulation, and Tooling](#)

Ian McCormack, Tomas Dougan, Sam Estep, Hanan Hibshi, Jonathan Aldrich, Joshua Sunshine.

ACM Transactions on Software Engineering and Methodology (TOSEM '24)

[A Study of Undefined Behavior Across Foreign Function Boundaries in Rust Libraries](#)

Ian McCormack, Joshua Sunshine, Jonathan Aldrich

IEEE/ACM International Conference on Software Engineering (ICSE '25)

[Gradual C0: Symbolic Execution for Efficient Gradual Verification](#)

Jenna DiVincenzo, **Ian McCormack**, Hemant Gouni, Jacob Gorenburg, Mona Zhang, Conrad Zimmerman, Joshua Sunshine, Éric Tanter, Jonathan Aldrich.

ACM Transactions on Programming Languages and Systems (TOPLAS '25)

[Plan B: Design Methodology for Cyber-Physical Systems Robust to Timing Failures](#)

Mohammad Khayatian, Mohammadreza Mehrabian, Edward Andert, Reese Grimsley, Kyle Liang, Yi Hu,

Ian McCormack, Carlee Joe-Wong, Jonathan Aldrich, Bob Iannucci, Aviral Shrivastava

ACM Transactions on Cyber-Physical Systems (TCPS '22)

[Computational Making via Bidirectional Parametric Modeling](#)

Chris Johnson and **Ian McCormack**.

Bridges '21

TALKS

Efficiently Finding Aliasing Bugs in Multilanguage Rust Applications - PhD Student Highlight Talk

Hanan Hibshi and **Ian McCormack** - April 29th, 2026

Centers of Academic Excellence in Cybersecurity - Community Symposium

[BorrowSanitizer: Finding Ownership Bugs in Multilanguage Rust Applications](#)

Ian McCormack - October 27th, 2025

2025 LLVM Developers Meeting - Workshop on Supporting Memory Safety in LLVM

[BorrowSanitizer](#)

Ian McCormack, Oliver Braunsdorf, Johannes Kinder, Jonathan Aldrich, Joshua Sunshine - May 5th, 2025

Rust Verification Workshop (RW2025) at ETAPS 2025

RESEARCH COMPETITIONS

TickTalk - One Program to Rule the Intersection - 🏆 (2nd)

Reese Grimsely, Edward Andert, **Ian C. McCormack**, Eve Hu, Bob Iannucci.

Student Design Competition: Networked Computing on the Edge.

Cyber-Physical Systems and Internet-of-Things Week (CPS-IoT Week '21)

A Software Library Model for the Internet of Things

Ian McCormack

ACM Student Research Competition (SPLASH '20)

EXPERIENCE

Research Assistant, Carnegie Mellon University's REUSE Program

June 2020 - May 2021

Mentors: [Bob Iannucci](#), Jonathan Aldrich

- Explored and prototyped a novel software repository system to allow combining disjointly-versioned modules.
- Implemented a networking layer for the [TTPython](#) distributed programming framework to support an autonomous vehicle intersection that won 2nd place in the 2021 CPS-IoT Week's Student Design Competition.

Technology Intern, Travelers Insurance, St. Paul, Minnesota

June - August, 2019

Mentor: [Mikko Niemioja](#)

- Solved ongoing issues with configuring production environments by creating a .NET application to allow partial backup and restoration of Windows Registry configurations.
- Identified employees' shared frustrations with onboarding and provisioning and used Angular to create a wireframe application that would improve the experience for new hires.